

IAN (HSINYEN) WU

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EDUCATION

Northeastern University

Sep 2022 – May 2025

Master of Science in Computer Science, GPA: 3.7/4.0

Boston, MA

- **Coursework:** Cloud Computing, Computer Networks, Database Design, Machine Learning, Web Development

WORK EXPERIENCE

Industrial Technology Research Institute (ITRI)

Aug 2025 – Present

Software Engineer

Remote

- Worked closely with clinical educators and trainees to design and integrate role-based AI agents into an **asynchronous multi-agent orchestration system** using **Python Django** and the **OpenAI SDK**, supporting realistic clinical simulation workflows used by **200+ trainees**.
- Built an **async evaluation pipeline** with dynamic agent routing and context compression, reducing response latency by **~65%**.
- Established automated **CI/CD** pipelines with **GitHub Actions** and SSH tunneling, enabling **35+ daily releases** and reducing deployment cycles from **~2h to ~8m**.

High-Frequency Trading Startup

Nov 2024 – Feb 2025

Software Engineer Intern

Chicago, IL

- Improved backtesting throughput and concurrency by prototyping a **distributed Master/Worker execution model** in **Rust**, replacing a monolithic ML simulation process.
- Split DAG-based simulations into independently schedulable tasks using **message queue(NATS)**, and added **asynchronous dispatch** with **leader election** to remove single-controller bottlenecks and enable safe controller failover.
- Standardized execution for CPU- and IO-bound ML simulation tasks by containerizing stateless workers with **Docker** and decoupling compute from shared data in **AWS S3**, enabling isolated and repeatable task execution without blocking the pipeline.

AI Roboto Edu

Jun 2024 – Sep 2024

Full Stack Engineer Intern

Los Angeles, CA

- Optimized **Java Spring Boot** backend queries for course listings and user progress tracking through composite indexing and query batching, reducing API response time from **~450ms to ~330ms** on key instructor and student-facing endpoints.
- Reduced form submission errors by **~35%** in course enrollment and quiz submission flows by implementing client-side validation in **React (Next.js)** using **Zod schemas** and real-time error feedback, reducing failed submissions in critical user workflows.
- Established unit and integration test coverage for enrollment, course management, and progress tracking modules using **JUnit** and **Mockito**, reducing regression risk during feature releases.

PROJECTS

SettleChat – AI-Powered Real-Time Chatroom (Go | WebSocket | NATS | PostgreSQL | Docker | LLMs)



- Built a real-time chat system using **WebSocket** and a **NATS-based event-driven architecture** for low-latency messaging.
- Implemented **LLM-based summarization** with token-efficient prompts, reducing context size by **~40%**.

Distributed Key-Value Storage System (Java | Raft Consensus | gRPC)



- Engineered a distributed key-value store in **Java** using the **Raft algorithm**, with segmented logs, snapshotting, and replication.
- Evaluated performance under controlled workloads, achieving **~14,000 QPS** with **~200ms P99 latency**.

TECHNICAL SKILLS

Programming Languages: Go, Java, Python, Rust, JavaScript, TypeScript, SQL, HTML/CSS

Frameworks & Libraries: React, Next.js, Express, FastAPI, Redux, Spring Boot

Databases & Storage: MySQL, PostgreSQL, Redis, MongoDB, Supabase

Infrastructure & Tools: Linux, AWS, GCP, Docker, Git, NATS, GitHub Actions, CI/CD